



University of
Inland Norway

ØKA2015 Applied Data Science

Autumn 2026 | 7.5 study points | Lillehammer

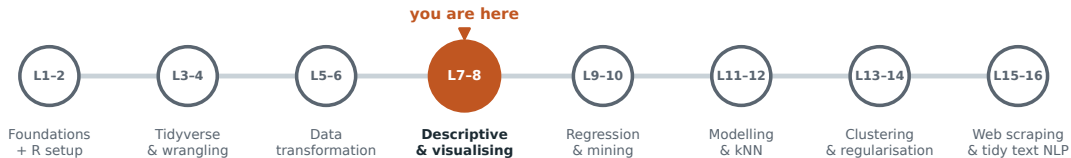
Lecture 7

Descriptive statistics and exploratory analysis

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Tuesday 6 October 2026 | 10:15–12:00

Where we are in the course



So far (Lectures 3 to 6)

Data into R; rows with `filter()` and `arrange()`; columns with `select()` and `mutate()`; groups with `group_by()` and `summarise()`; missing values filled.

Today

You *describe* data with numbers: what kind of data it is, where its centre is, how spread out it is, which values are unusual, and how two variables move together.

Materials for this lecture

The work is done in **RStudio on your own laptop**, typing each command yourself. Everything below is also online.

Slides | this document, as a PDF

R script | `lecture07_descriptive.R`: every command from today, in today's order, each line explained in a comment. We run it in parallel.

R Markdown tutorial | `lecture07_descriptive.Rmd` and its knitted PDF: the same content as a document

Practice exercise | short tasks on the housing data, not graded

Data | `flights` and `weather`, both inside the `nycflights13` package

Primary: RStudio on your laptop. GitHub and Colab are for afterwards, to download or run the code.

Where it is

GitHub, the course repository:

[github.com/touseefhameed/
oka2015-applied-data-science](https://github.com/touseefhameed/oka2015-applied-data-science)

Google Colab, the script in your browser, nothing to install: [Open in Colab](#), also a badge on the GitHub page



Reading

Camm et al., *Business Analytics*, Ch. 2
Descriptive Statistics. Wright et al., *Tidyverse Skills for Data Science*, Ch. 5.4 and 5.5.

Today's plan

- 1 **Types of data:** population and sample, quantitative and categorical, cross-section and time series
- 2 **Distributions:** frequency tables, histograms, shape
- 3 **Central tendency:** mean, median, mode
- 4 **Variability:** variance, standard deviation, quartiles
- 5 **Outliers:** the IQR rule, boxplots, z-scores
- 6 **Association:** scatter plots, covariance, correlation

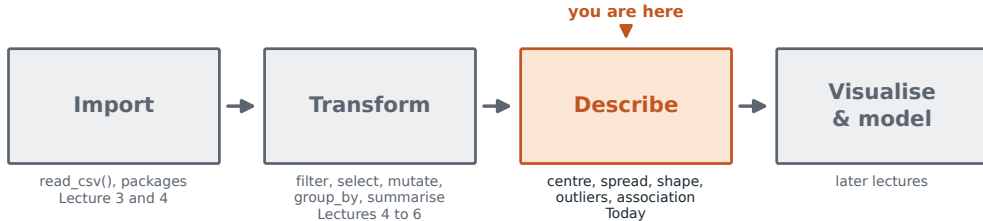
How the session runs

Each code slide is one block of the script. Type it, run it, read the output. Two short “Your turn” slides let you try it alone.

By the end you will be able to

- ✓ say what type a variable and a dataset are
- ✓ describe the shape of a distribution
- ✓ report the centre and the spread, and choose the right measure
- ✓ flag outliers and decide what to do with them
- ✓ measure how two variables move together

1. Describing data: the words we need



Data | facts and figures collected, analysed and summarised for presentation and interpretation.

Variable | a characteristic that can take different values: a column.

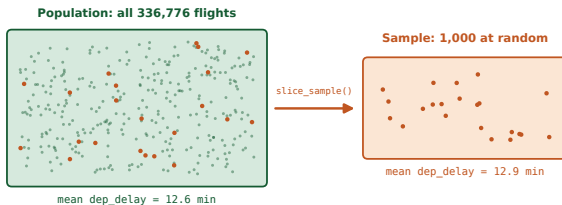
Observation | the set of values for one unit: a row.

Camm et al., *Business Analytics*, Ch. 2.1

Key idea

Descriptive statistics summarise data with a few numbers and pictures. The aim is to understand the **variation**: how a variable differs from one observation to the next.

1. Population and sample



```
set.seed(2026)
flights_sample <- flights |>
  slice_sample(n = 1000)
mean(flights$dep_delay, na.rm = TRUE)
mean(flights_sample$dep_delay, na.rm = TRUE)
```

```
[1] 12.63907
[1] 12.89673
```

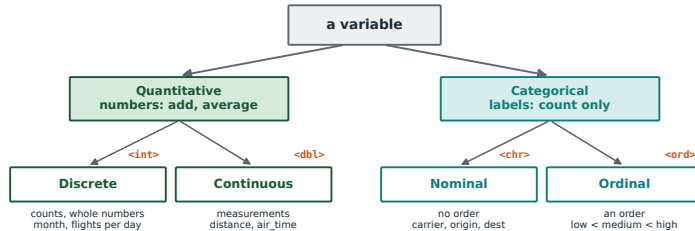
Population | every element of interest: all 2013 flights from New York.

Sample | a subset of the population. A random sample is representative: close to the truth on average.

set.seed() | fixes the random draw, so everyone gets the same 1,000 flights.

Camm et al., Ch. 2.2. In practice we nearly always work with samples.

1. Quantitative and categorical variables



```
flights |>
  select(carrier, month, dep_delay) |>
  glimpse()
```

```
$ carrier <chr> "UA", "UA", "AA", ...
$ month <int> 1, 1, 1, 1, 1, ...
$ dep_delay <dbl> 2, 4, 2, -1, -6, ...
```

Quantitative | you can add and average it.
<int>, <dbl>.

Categorical | you can only count it. <chr>,
<fct>.

Careful | a number can be a label: a post-code, a 1 to 5 rating.

1. Cross-sectional and time series data

Cross-section: 1 January, many flights

carrier	dest	dep_delay
UA	IAH	2
AA	MIA	2
B6	BQN	-1
DL	ATL	-6
...	...	...

842 rows, one point in time



```
flights |> filter(month == 1, day == 1)
flights |> count(month)
```

```
# A tibble: 842 x 19
# A tibble: 12 x 2
```

Cross-sectional | many units at about the same point in time.

Time series | one quantity measured over many time periods. Used to see trends and to forecast.

Camm et al., Ch. 2.2

2. Descriptive and exploratory analysis

Descriptive analysis

Summarises what is in the data: centre, spread, shape. It describes. It does not explain or predict.

Example: the average delay of New York flights in 2013, overall and per airport.

Exploratory analysis

Looks for patterns and relationships you did not know about: which variables move together, which groups differ.

Example: are flights that leave late also the ones that arrive late?

Wright et al., *Tidyverse Skills for Data Science*, Ch. 5.4 and 5.5

Steps of an exploratory analysis

- ① look for missing values
- ② look for outliers
- ③ calculate summary statistics
- ④ use plots to see shape and patterns
- ⑤ look at relationships between variables

Key idea

Exploration suggests questions. It does not prove answers. **Correlation is not causation.**

2. A distribution of a categorical variable: a frequency table

```
flights |>
  count(carrier, sort = TRUE) |>
  mutate(percent = round(100 * n / sum(n), 1))
```

```
# A tibble: 16 x 3
  carrier n percent
<chr> <int> <dbl>
1 UA  58665  17.4
2 B6  54635  16.2
3 EV  54173  16.1
4 DL  48110  14.3
```

Distribution | which values a variable takes, and how often.

Frequency | the count per category: `n`.

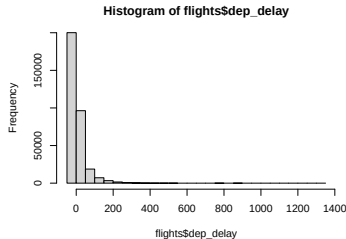
Percent frequency | the share of all observations: $100 * n / \text{sum}(n)$. United flew 17.4% of the flights.

Tools | `count()` from Lecture 4, `mutate()` from Lecture 5. Nothing new.

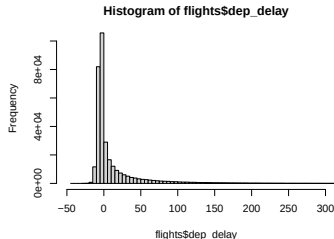
Camm et al., Ch. 2.4 *Frequency distributions for categorical data*

2. A distribution of a quantitative variable: a histogram

```
hist(flights$dep_delay)
```



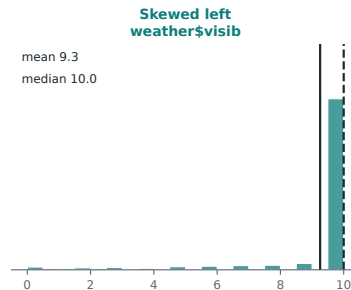
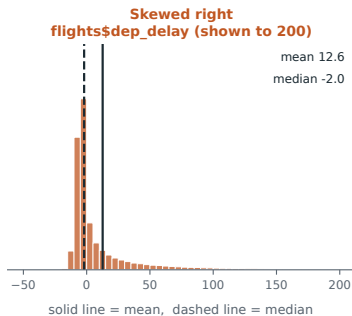
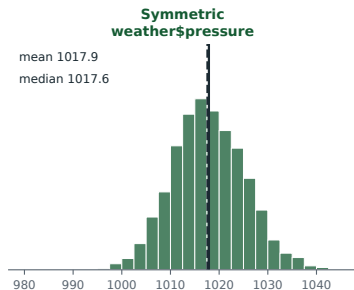
```
hist(flights$dep_delay, breaks = 200,  
     xlim = c(-50, 300))
```



Histogram | the range split into bins, one bar per bin, height = how many values fall in it. **breaks** sets the number of bins, **xlim** the part of the axis shown.

Camm et al., Ch. 2.4 *Histograms*

2. The shape of a distribution



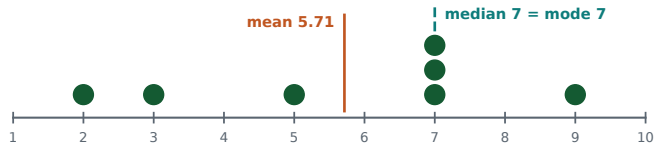
Symmetric | the left side mirrors the right. Mean $\approx$ median.

Skewed right | a long tail to the right. Mean $>$ median. Incomes, delays, prices.

Skewed left | a long tail to the left. Mean $<$ median. Easy exam scores.

Camm et al., Ch. 2.4. A bell-shaped symmetric distribution is called *normal*.

3. Central tendency: mean, median, mode



```
books <- c(3, 7, 2, 9, 7, 5, 7)
mean(books)
median(books)
tibble(books) |> count(books, sort = TRUE)
```

```
[1] 5.714286
[1] 7
  books n
1 7 3
```

Mean | the sum divided by the count: $\bar{x} = \frac{\sum x_i}{n} = \frac{40}{7}$.

Median | the middle value once sorted: 2 3 5 **7** 7 7 9.

Mode | the most frequent value. R has no function for it, but `count(sort = TRUE)` puts it on top.

Camm et al., Ch. 2.5 *Measures of location*

3. Mean or median? Real data decides

```
mean(flights$dep_delay, na.rm = TRUE)
median(flights$dep_delay, na.rm = TRUE)
flights |> count(dep_delay, sort = TRUE)
```

```
[1] 12.63907
[1] -2
  dep_delay n
1 -5 24821
```

Mean 12.6 | pulled up by a few very late flights. One flight of 1,301 minutes adds as much as 100 flights of 13 minutes.

Median -2 | the typical flight left two minutes *early*.

Mode -5 | the single most common delay.

Key idea

The mean uses every value, so extreme values pull it. The median uses only the order, so they do not. For skewed data, report the **median**, or both.

3. The centre for each group

```
flights |>
  group_by(origin) |>
  summarise(
    mean = mean(dep_delay, na.rm = TRUE),
    median = median(dep_delay, na.rm = TRUE),
    n = n())
```

```
origin mean median n
1 EWR 15.108 -1 120835
2 JFK 12.112 -1 111279
3 LGA 10.347 -3 104662
```

Grouped description | the same measures for each category. `group_by()` and `summarise()` from Lecture 6.

Reading it | Newark has the highest mean delay. All three medians are below zero: at every airport the typical flight leaves on time.

Always add n | a mean of 5 values and a mean of 100,000 do not deserve the same trust.

3. Filling a missing value: mean, median or mode?

You filled missing values in Lectures 5 and 6. The shape decides which value to use:

variable	fill with
quantitative, symmetric	mean
quantitative, skewed or with outliers	median
categorical	mode

Filling changes the data. Do it only with a reason, and say so in your report.

```
weather_filled <- weather |>
  mutate(pressure = replace_na(pressure,
                                mean(pressure, na.rm = TRUE)))
sum(is.na(weather$pressure))
sum(is.na(weather_filled$pressure))
```

```
[1] 2729
[1] 0
```

```
flights |> count(carrier, sort = TRUE) |>
  slice(1)
```

```
1 UA 58665
```

`pressure` is symmetric, so the mean is fine. `dep_delay` is skewed: fill it with the median.

4. Variability: variance and standard deviation

```
scores <- c(70, 75, 80, 85, 90)
tibble(scores) |>
  mutate(deviation = scores - mean(scores),
         squared = deviation^2)
var(scores)
sd(scores)
```

```
  scores deviation squared
1  70      -10     100
2  75       -5      25
3  80       0       0 ...
[1] 62.5
[1] 7.905694
```

Variance | the average squared distance from the mean:

$$s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} = \frac{250}{4} = 62.5$$

Why squared | so that minus and plus distances do not cancel.

Why $n-1$ | for a sample it gives a better estimate of the population variance.

Standard deviation | $s = \sqrt{s^2} = 7.9$. Back in the units of the data: points. This is the one we report.

Camm et al., Ch. 2.6 *Measures of variability*

4. Spread on real data: sd, quartiles, IQR

```
sd(flights$dep_delay, na.rm = TRUE)
range(flights$dep_delay, na.rm = TRUE)
quantile(flights$dep_delay, c(0.25, 0.50,
                             0.75),
         na.rm = TRUE)
IQR(flights$dep_delay, na.rm = TRUE)
summary(flights$dep_delay)
```

```
[1] 40.21006
[1] -43 1301
25% 50% 75%
-5 -2 11
[1] 16
   Min. 1st Qu. Median Mean 3rd Qu. Max.
-43.00 -5.00 -2.00 12.64 11.00 1301.00
```

Range | max minus min. Uses two values only, so one extreme value decides it.

Percentile | the p th percentile has $p\%$ of the values below it.

Quartiles | Q1 = 25th, Q2 = median, Q3 = 75th percentile.

IQR | $Q3 - Q1 = 16$ minutes: the spread of the middle 50%. Not moved by extreme values.

Coefficient of variation | sd / mean : spread relative to the mean, without units.

Camm et al., Ch. 2.6 and 2.7. `summary()` gives the five-number summary and the mean in one line.

Your turn 1: centre and spread

Use `flights$arr_delay`, the arrival delay.

- ① What are the mean and the median?
Which way is it skewed?
- ② What is the most common arrival delay?
- ③ What are the sd and the IQR?
- ④ Which origin airport has the highest median arrival delay?

Remember `na.rm = TRUE`.

Your turn 1: centre and spread

Use `flights$arr_delay`, the arrival delay.

- ① What are the mean and the median?
Which way is it skewed?
- ② What is the most common arrival delay?
- ③ What are the sd and the IQR?
- ④ Which origin airport has the highest median arrival delay?

Remember `na.rm = TRUE`.

Answers

```
mean(flights$arr_delay, na.rm = TRUE) # 6.9
median(flights$arr_delay, na.rm = TRUE) # -5
flights |> filter(!is.na(arr_delay)) |>
  count(arr_delay, sort = TRUE) # -13
sd(flights$arr_delay, na.rm = TRUE) # 44.6
IQR(flights$arr_delay, na.rm = TRUE) # 31
flights |>
  group_by(origin) |>
  summarise(m = median(arr_delay,
                        na.rm = TRUE))
```

Mean > median: skewed right. Without the filter, `count()` puts the NA row first. EWR has the highest median, -4.

5. Outliers

Values far from the rest

Find them, explain them, then decide

5. What an outlier is, and what to do with it

Outlier | an observation that lies far from the other observations.

Why it matters | one outlier can move the mean, the sd and the correlation a long way.

Two causes

- **An error**: a typing mistake, a broken sensor. A wind speed of 1,048 mph.
- **A real but rare case**: a flight 1,301 minutes late after a long breakdown.

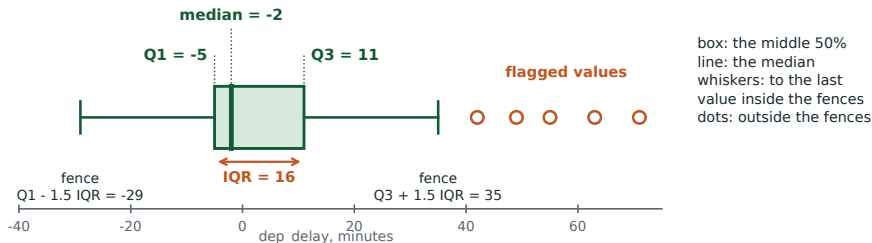
The rule

Never remove an observation without a reason.

- ① find it with a rule or a plot
- ② check where it comes from
- ③ a confirmed error: correct or remove it, and write down what you did
- ④ a real value: keep it, and use measures it does not distort (median, IQR)

Camm et al., Ch. 2.7 *Identifying outliers*; Wright et al., Ch. 5.4

5. The IQR rule and the boxplot



```
boxplot(flights$dep_delay, ylim = c(-50, 150))
flights |>
  filter(dep_delay < -29 | dep_delay > 35) |>
  nrow()
```

```
[1] 43216
```

Fences | $Q1 - 1.5 IQR$ and $Q3 + 1.5 IQR$.
Values outside are flagged.

43,216 flagged | 13% of flights. In skewed data the rule flags many real values. A flag is not a verdict.

5. The z-score rule

```
m <- mean(flights$dep_delay, na.rm = TRUE)
s <- sd(flights$dep_delay, na.rm = TRUE)
flights |>
  mutate(z = (dep_delay - m) / s) |>
  filter(abs(z) > 3) |>
  nrow()
```

```
[1] 7928
```

First store the mean and sd as `m` and `s`. `abs()` removes the sign, so both tails count.

Camm et al., Ch. 2.7 *z-scores*, *Empirical rule*

z-score | $z = \frac{x - \bar{x}}{s}$: how many standard deviations a value lies from the mean. 0 is the mean; -2 is two sd below it.

Flag | $|z| > 3$.

Empirical rule | for a bell-shaped distribution, about 68% of values lie within 1 sd, 95% within 2, and nearly all within 3. So $|z| > 3$ is rare.

Best for | symmetric data. For skewed data, prefer the IQR rule.

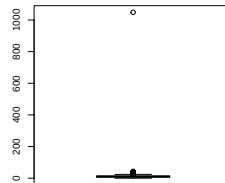
5. A real data entry error

```
weather |>
  arrange(desc(wind_speed)) |>
  select(origin, month, day, hour, wind_speed)
```

```
origin month day hour wind_speed
1 EWR 2 12 3 1048.4
2 EWR 1 31 6 42.6
```

```
weather_clean <- weather |>
  filter(wind_speed < 200 | is.na(wind_speed))
nrow(weather) - nrow(weather_clean)
```

```
[1] 1
```



1,048 mph | a hurricane is about 75 mph. This is a recording error: remove it, and write it down.

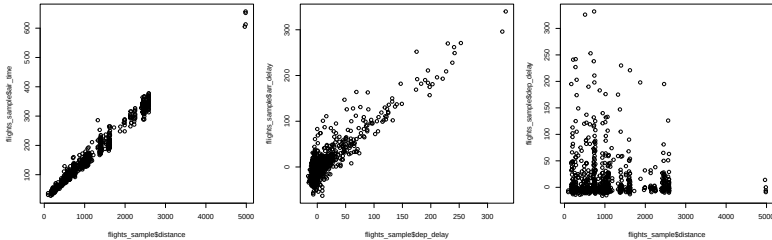
1,301 minutes | a real flight. Keep it.

6. Two variables

Scatter plots, covariance, correlation

Do they move together?

6. The scatter plot



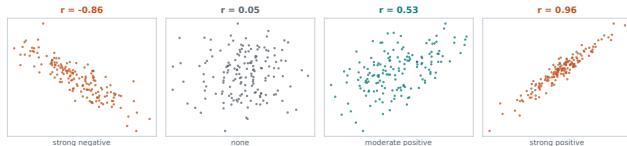
```
plot(flights_sample$distance, flights_sample$arr_time)
plot(flights_sample$dep_delay, flights_sample$arr_delay)
plot(flights_sample$distance, flights_sample$dep_delay)
```

Scatter plot | one dot per observation, x against y.

Left to right | tight and upward; upward; no pattern.

Camm et al., Ch. 2.8 *Scatter charts*. The 1,000-flight sample keeps the plot fast.

6. Covariance and correlation



```
cov(flights$distance, flights$air_time,  
     use = "complete.obs")  
cor(flights$distance, flights$air_time,  
     use = "complete.obs")  
cor(flights$distance, flights$dep_delay,  
     use = "complete.obs")
```

```
[1] 68301.35  
[1] 0.9906496  
[1] -0.02167079
```

Covariance | positive: they move together; negative: they move apart. Its size depends on the units: in km it is 109,897.

Correlation r | covariance rescaled to -1 to $+1$, with no units. 0 means no linear relationship.

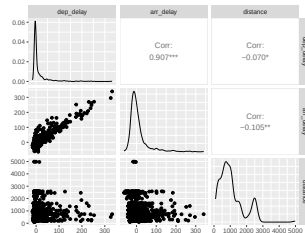
`use = "complete.obs"` | skip rows with a missing value.

6. Many pairs at once, and a warning

```
flights |>
  select(dep_delay, arr_delay,
         air_time, distance) |>
  cor(use = "complete.obs") |>
  round(2)
```

```
      dep_delay arr_delay air_time distance
dep_delay 1.00  0.91 -0.02 -0.02
arr_delay 0.91  1.00 -0.04 -0.06
air_time -0.02 -0.04 1.00  0.99
distance -0.02 -0.06 0.99  1.00
```

```
flights_sample |>
  select(dep_delay, arr_delay, distance) |>
  ggpairs()
```



`ggpairs()` | from GGally: scatter plots, densities and r for every pair.

Correlation is not causation. $r = 0.91$ says late departures and late arrivals go together. It does not say why.

Your turn 2: outliers and association

Use the `weather` table.

- ① Draw a boxplot of `temp`. Any flagged values?
- ② How many hours have a `humid` value more than 3 sd from the mean?
- ③ What is the correlation between `temp` and `dewp` (dew point)? And between `temp` and `humid`?

Your turn 2: outliers and association

Use the `weather` table.

- 1 Draw a boxplot of `temp`. Any flagged values?
- 2 How many hours have a `humid` value more than 3 sd from the mean?
- 3 What is the correlation between `temp` and `dewp` (dew point)? And between `temp` and `humid`?

Answers

```
boxplot(weather$temp) # none flagged
m <- mean(weather$humid, na.rm = TRUE)
s <- sd(weather$humid, na.rm = TRUE)
weather |> mutate(z = (humid - m) / s) |>
  filter(abs(z) > 3) |> nrow() # 0
cor(weather$temp, weather$dewp,
     use = "complete.obs") # 0.89
cor(weather$temp, weather$humid,
     use = "complete.obs") # 0.08
```

Warm air holds more water, so temperature and dew point move together. Humidity is a share, so it barely moves with temperature.

Check your understanding

1. House prices: mean 250,000, median 180,000.

Which way are they skewed?

2. Why do we report the standard deviation, not the variance?

3. $Q1 = 10$ and $Q3 = 30$. Where are the IQR fences?

4. A survey records satisfaction as 1, 2, 3, 4, 5.

Quantitative or categorical?

5. Two columns have $r = -0.85$. What does the scatter plot look like?

Check your understanding

1. House prices: mean 250,000, median 180,000. Which way are they skewed?
2. Why do we report the standard deviation, not the variance?
3. $Q1 = 10$ and $Q3 = 30$. Where are the IQR fences?
4. A survey records satisfaction as 1, 2, 3, 4, 5. Quantitative or categorical?
5. Two columns have $r = -0.85$. What does the scatter plot look like?

Answers

1. Right: mean $>$ median. A few very expensive houses pull the mean up.
2. The sd is in the units of the data; the variance is in squared units.
3. IQR = 20. Fences: $10 - 30 = -20$ and $30 + 30 = 60$.
4. Categorical, ordinal: an order, but the steps need not be equal.
5. A clear downward cloud: as one rises, the other falls.

Course materials online

Everything from this lecture is online, and stays there for the whole course:

GitHub | github.com/touseefhameed/oka2015-applied-data-science

the slides, the R script, the R Markdown document and its PDF, and the practice exercise, in the `lecture07` folder.

Google Colab | [Open in Colab](#), or the Lecture 7 badge on the GitHub page
the same script, running in your browser. No installation.

Colab in three steps

Open the link → sign in with any Google account → press the play button on the first cell. It installs the packages for you, about two minutes.

Primary and secondary

Primary: RStudio on your own laptop, typing the commands yourself.

Afterwards: download the script from GitHub to rerun it, or run it directly in Colab with nothing installed.

Scan with your phone



GitHub



Colab, Lecture 7

What we covered today

Concepts

- ✓ population and sample; quantitative and categorical; cross-section and time series
- ✓ the shape of a distribution: symmetric, skewed right, skewed left
- ✓ mean, median, mode, and when each is the right one
- ✓ variance, sd, quartiles, IQR
- ✓ outliers: flag, check, then decide
- ✓ correlation is not causation

Tools

- ✓ `slice_sample()`, `set.seed()`
- ✓ `count()` with percentages; `hist()`
- ✓ `mean()`, `median()`, `sd()`, `var()`, `range()`
- ✓ `quantile()`, `IQR()`, `summary()`
- ✓ `boxplot()`, z-scores with `mutate()` and `abs()`
- ✓ `plot()`, `cov()`, `cor()`, `ggpairs()`

If you can describe a new variable with its type, shape, centre, spread and outliers, today's objective is met.

Before the next lecture

To do

- ✓ finish the script if you did not
- ✓ the practice exercise: the same steps on the housing data from Assignment 1
- ✓ read Camm et al., Ch. 2, sections 2.1, 2.2 and 2.4 to 2.8

Next lecture

Thursday 8 October.

The thread

Lectures 3 to 6: get data in, and change it.
Today: describe it with numbers.

Sources used in this lecture

Course reading

- Camm, J. D., Fry, M. J., Cochran, J. J. & Ohlmann, J. W. (2024). *Business Analytics*. Cengage. Ch. 2 *Descriptive Statistics*: types of data, distributions, measures of location and variability, analysing distributions, measures of association.
- Wright, C., Ellis, S., Hicks, S. & Peng, R. (2021). *Tidyverse Skills for Data Science in R*, Ch. 5.4 *Descriptive analysis* and 5.5 *Exploratory analysis*.
<https://jhudatascience.org/tidyversecourse/>

Packages and data

- nycflights13: Wickham, H. (2021). Flights and weather for New York airports in 2013. US Bureau of Transportation Statistics. <https://github.com/tidyverse/nycflights13>
- GGally: Schloerke, B. et al. Extension to ggplot2. <https://ggobi.github.io/ggally/>

Course material

- GitHub: <https://github.com/touseefhameed/oka2015-applied-data-science> Google Colab: [Open in Colab](#)

Thank you



Course materials: github.com/touseefhameed/oka2015-applied-data-science

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